

TEJES .K. IYER

Robotics Engineer — Humanoid Robots | DRL | ROS 2 | Motion Planning | Physical AI

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SUMMARY

Mechatronics Engineering graduate with end-to-end experience building real robotic systems — from URDF/XACRO modelling and ROS 2 control to DRL-based locomotion, manipulation, and computer vision. Built a bipedal humanoid robot and a 6-DOF robotic arm with DRL trajectory control entirely from scratch. Research internship at IIT Tirupati (Dept. of Science & Technology, Govt. of India). Passionate about humanoid robots, physical AI, and sim-to-real deployment.

EDUCATION

Chandigarh University — B.E. Mechatronics Engineering

July 2022 – June 2026

Relevant: Robotics, Control Systems, Embedded Systems, Machine Learning, Mechatronic System Design

Blossom Public School — Tamil Nadu, India

2021

Class XII: 78% | Class X: 76%

TECHNICAL SKILLS

Robotics & Control: ROS 2 (Humble), MoveIt 2, Nav2, URDF/XACRO, TF2, Inverse Kinematics, PID Control, Sensor Fusion, SLAM

DRL & AI: PPO, DDPG, Stable-Baselines3, PyTorch, TensorFlow Lite, Sim-to-Real Transfer, Reward Shaping

Simulation: Gazebo, NVIDIA Isaac Sim (learning), PyBullet

Vision & Perception: OpenCV, YOLO (object detection), Pi AI Camera, ROS 2 vision pipeline integration

Programming: Python, C++, Linux, Git, Docker

Hardware: Raspberry Pi 5, Arduino, IMU Sensors, Servo Actuators, LoRa/LoRaWAN, LiDAR

CAD: SolidWorks, Fusion 360, AutoCAD

WORK EXPERIENCE

Research Intern | IIT Tirupati (IITTNiF) — Dept. of Science & Technology, Govt. of India

Jun – Aug 2024

- Field-deployed a LoRaWAN GPS tracking system (Raspberry Pi + SX1278 LoRa chip) — hardware setup, firmware programming, and on-site testing across urban and open environments achieving 2+ km range.
- Benchmarked RSSI and packet loss; delivered technical report to IITTNiF.

PROJECTS

Humanoid Bipedal Robot — DRL Locomotion & Sim-to-Real Transfer

Nov 2025 – Jun 2026

- Modelled robot in URDF/XACRO for ROS 2; 19-servo inverse kinematics controller with PID-based balance. Trained PPO agent (Stable-Baselines3) in Gazebo with multi-objective reward shaping — stable walking within 2M timesteps.
- Implementing YOLO-based vision navigation (Raspberry Pi 5 + Pi AI Camera) and actively working on sim-to-real transfer — deploying learned gait policy onto physical hardware.
- Extended locomotion training to MuJoCo with SAC (Stable-Baselines3) for faster iteration and integrated IMU/FSR sensor fusion for balance feedback.

6-DOF Robotic Arm — Manipulation, Motion Planning & DRL Control

Aug – Dec 2024

- Modelled in URDF/XACRO for ROS 2 + MoveIt 2; collision-aware motion planning. DDPG-based DRL trajectory controller as alternative to classical IK. Real-time YOLO pick-and-place. Sub-10mm positional repeatability.

CERTIFICATIONS

- STAR PNT Program — LoRa WAN Based GPS Tracking System | IIT Tirupati Navavishkar I-Hub Foundation | Dept. of Science & Technology, Govt. of India | 2024